

Chapter 90 — Fundamentals of Nephrology

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INTRODUCTION

- **Why urologists need renal physiology:** formal teaching is largely preclinical and the rest is learnt on the job; nephrology cover is available, but four reasons justify retaining it.
- **Early fluid and electrolyte management** — the urologist primarily in charge who starts correct fluid, electrolyte and perioperative treatment early may secure a better outcome than one who waits for a consultant.
- **Favourable perioperative outcomes** — knowing which perioperative factors impair renal function allows effective collaboration with anaesthetists to minimise kidney-related complications.
- **Optimal management of kidney disease** — pathophysiology informs the choice of surgical approach, perioperative care and postoperative management.
- **Collaborative care** — a shared physiological language with nephrologists, oncologists and anaesthetists improves multidisciplinary care.

Concept — *The kidney is the organ of homeostasis, so a urologist who operates on it, obstructs it, relieves it or doses drugs through it is altering volume, electrolytes and acid-base balance whether or not a nephrologist is present.*

OVERVIEW OF NORMAL STRUCTURE AND FUNCTION OF THE KIDNEY

The Nephron

- **About 1 million nephrons per kidney**, tightly packed alongside blood vessels; each is a **glomerulus** (filters fluid and small molecules from blood) plus its paired **tubule**.
- **Tubule sequence:** receives glomerular filtrate → **reabsorbs the majority of filtered molecules** → secretes excess metabolic waste into the filtrate → concentrates it into final urine; together the nephrons of one kidney excrete about **1–2 L of urine per day**.
- **Nephron number is fixed at birth** and declines with age from about 25 years.
- **Reduced nephron mass → hyperfiltration:** after nephrectomy or with unilateral renal agenesis, remaining nephrons increase single-nephron filtration to maintain glomerular filtration and homeostasis.
- **Glomerulus:** a tangle of capillaries lined internally by a single layer of **endothelial cells** and covered externally by **podocytes** (visceral epithelial cells) — highly differentiated cells whose **foot processes** interdigitate to form **filtration slits** bridged by **slit diaphragms**.
- **Glomerular basement membrane (GBM):** a collagen-based layer produced by endothelial cells, lying between endothelium and podocyte.
- **Filtration barrier = endothelium + GBM + podocyte;** it regulates passage from blood to urine by **size and charge**.
- **Mesangium** — a stalk of specialised cells and matrix that holds the capillary tuft together.
- **Bowman's space:** the capillary tuft invaginates into the initial tubule; this space is lined by **parietal epithelium**, whereas the podocytes (visceral epithelium) form the barrier. Filtrate then passes proximal tubule → loop of Henle → distal tubule → collecting duct.
- **Glomerular landmarks (Fig. 90.2):** afferent arteriole bearing **juxtaglomerular granular cells** and sympathetic nerve endings; efferent arteriole; **macula densa** of the distal convoluted tubule; extraglomerular mesangial cells; urinary space draining at the **urinary pole** into the proximal tubule.

Renal Structure and Blood Supply

- **Four zones:** cortex, outer medulla (outer stripe and inner stripe) and inner medulla. Fig. 90.1 maps arcuate and interlobular arteries, vasa recta, loop limbs and cortical, outer and inner medullary collecting ducts.
- **All glomeruli lie in the cortex**, but the pair differs in tubule length:
 - **Cortical glomeruli** — short tubules that do not travel deep into the medulla.
 - **Juxtamedullary glomeruli** — long tubules descending deep into the inner medulla; this matters for medullary perfusion and ischaemic injury.

Basic Renal Homeostatic Functions

- **Excretion of waste and exogenous substances** — nitrogenous waste from hepatic metabolism; drugs, contrast and other ingested products or their breakdown products.
- **Blood pressure and volume — sodium is the currency of volume:** sodium retention raises blood volume, excretion lowers it.
 - **Renin-angiotensin-aldosterone system (RAAS) sequence:** low renal perfusion pressure or sympathetic activation → renin from specialised renal cells → **angiotensin II** (potent vasoconstrictor, also directly increases renal sodium reabsorption) → adrenal **aldosterone** → distal sodium reabsorption → volume retention.
- **Acid-base, water and electrolyte balance** between intracellular and extracellular compartments, held within a tight range for cellular function.
- **Erythropoietin** — produced by **renal interstitial fibroblasts in response to hypoxia**; drives red cell maturation in the marrow.
- **Calcitriol (1,25-dihydroxyvitamin D, activated vitamin D)** — hydroxylation of 25-hydroxyvitamin D in **proximal tubule cells**, stimulated by raised parathyroid hormone or low calcium or phosphate; essential for calcium-phosphate homeostasis and bone.

Concept — *Structure dictates vulnerability: the glomerulus is a pressure-driven filter whose output is set by arteriolar tone, and the tubule is an energy-hungry reclamation line whose deepest segments work at the edge of hypoxia; almost every disorder in this chapter is a failure of one or the other.*

MEASUREMENT OF KIDNEY FUNCTION

- **Two settings, two approaches: steady state** (function stable over time, as at a routine clinic visit) versus **dynamic state** (acute fluctuation, as postoperatively).

Measurement of Kidney Function in Steady State

- **Glomerular filtration rate (GFR) is the standard index** of kidney function, but it cannot be measured directly.
- **Gold standard: serial urinary or plasma clearance of an exogenous marker** (inulin, iothexol, iothalamate) — reliable but cumbersome and costly, so surrogates are used in practice; the most widely used is **serum creatinine (SCr)**.

Serum Creatinine-Based Measures of Kidney Function

- **Creatinine is a muscle breakdown product, freely filtered** at the glomerulus — hence a good filtration marker.
- **Small active tubular secretion** from peritubular capillaries into urine is negligible in health but becomes an **increasing share of creatinine excretion as function deteriorates**.

- **24-hour creatinine clearance** (SCr, 24-hour urine volume and urine creatinine → mL/min) is little used: the collection is cumbersome and prone to under- or over-collection, and because of secretion it **overestimates GFR increasingly as renal function declines**.
- **SCr alone is insufficient** because creatinine generation varies independently of kidney function — **muscle mass is the most important factor**, varying with age, sex and race (Table 90.1).
- **Estimating equations** were derived in large, diverse populations with measured GFR, simultaneous creatinine and demographic data, incorporating age, sex and, historically, race.
- **Problem with a race coefficient:** it raised estimated GFR (eGFR) in Black patients, affecting chronic kidney disease (CKD) staging and drug dosing and potentially **delaying nephrology or transplant referral**.
- **Current recommendation** (American Society of Nephrology and National Kidney Foundation Task Force): laboratories report the **2021 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation, using age and sex without race** — derived in a diverse cohort and meeting performance standards without disadvantaging any population. Online calculators make memorisation unnecessary.

TABLE 90.1 SOURCES OF ERROR IN INTERPRETATION OF SERUM CREATININE AND SERUM CYSTATIN C

Factor	Serum creatinine	Serum cystatin C
Generation increased	Large muscle mass, high-protein diet, cooked meat, creatine supplements	Hyperthyroidism, glucocorticoid excess, obesity, inflammation, smoking
Generation decreased	Small muscle mass, limb amputation, muscle-wasting disease	Hypothyroidism
Tubular secretion decreased (level rises)	Drug inhibitors of secretion: trimethoprim, cimetidine, fenofibrate	Not applicable
Extrarenal elimination decreased (level rises)	Urine leak with peritoneal absorption; antibiotic inhibition of gut creatininase	—
Extrarenal loss increased (level falls)	Dialysis; large extracellular fluid losses (pleural or ascitic drainage)	Large extracellular fluid losses (pleural or ascitic drainage)

Modified from Gilbert SF, Weiner DE, Bombardier AS, et al., *National Kidney Foundation (NKF) Primer on Kidney Diseases*, 8th ed, 2022, Table 3.2.

- **What the table is for:** every row is a way the marker moves **without GFR changing**.
- **Pattern:** creatinine is confounded by muscle and diet; cystatin C by thyroid state, steroids and inflammation — the two fail in different patients, which is why combining them improves accuracy.
- **Traps:** SCr rise on **trimethoprim or cimetidine** is blocked secretion, not injury; **urine leak with peritoneal reabsorption** raises SCr without renal injury; SCr falls after large-volume drainage without better function.

Serum Cystatin C–Based Measures of Kidney Function

- **Cystatin C:** a 13-kDa protein made by all nucleated cells, freely filtered, then **completely reabsorbed and degraded by the proximal tubule** — so there should be little or none in urine in health.
- **Not wholly GFR-specific:** some non-GFR influences exist (Table 90.1).
- **Cystatin C equations require age and sex adjustment but have never shown race dependence.**
- **Combined creatinine–cystatin C 2021 equation (without race) is more accurate** than the creatinine-only equation, with smaller differences between race groups.
- **Indications:** an adjunct to stage function when **creatinine-based eGFR is likely biased** (Table 90.1) or when **staging sits at a consequential threshold** and confirmation is needed.
- **Barriers:** standardisation to a reference assay, higher cost than creatinine, clinician unfamiliarity — all expected to ease with a central reference standard, better reimbursement and experience.

TABLE 90.2 GFR-ESTIMATING EQUATIONS BASED ON SERUM CREATININE AND CYSTATIN C (2021 CKD-EPI; ML/MIN/1.73 M²)

Equation	Formula	Constants
Creatinine	$142 \times \min(\text{SCr}/\kappa, 1)^\alpha \times \max(\text{SCr}/\kappa, 1)^{-1.200} \times 0.9938^{\text{age}} \times 1.012$ if female	κ 0.7 female, 0.9 male; α -0.241 female, -0.302 male
Cystatin C	$133 \times \min(\text{Scys}/0.8, 1)^{-0.499} \times \max(\text{Scys}/0.8, 1)^{-1.328} \times 0.996^{\text{age}} \times 0.932$ if female	No sex-specific κ or α
Creatinine–cystatin C	$135 \times \min(\text{SCr}/\kappa, 1)^\alpha \times \max(\text{SCr}/\kappa, 1)^{-0.544} \times \min(\text{Scys}/0.8, 1)^{-0.323} \times \max(\text{Scys}/0.8, 1)^{-0.778} \times 0.9961^{\text{age}} \times 0.963$ if female	κ 0.7 female, 0.9 male; α -0.219 female, -0.144 male

min = lesser of the ratio or 1; *max* = greater of the ratio or 1; Scys = serum cystatin C. Gilbert et al., NKF Primer, 8th ed, 2022, Table 3.1.

- **Pattern:** all three use **age and sex**; **none uses race**; each applies an age decay and a sex multiplier.
- **Trap:** the female multiplier is **above 1 in the creatinine equation** but **below 1 in the cystatin C and combined equations**; the formulae need not be memorised.

Urinary Protein Excretion

- **Two defences against protein loss:** the filtration barrier limits passage of albumin and non-albumin protein, and the **proximal tubule reclaims protein** that enters filtrate → **normal excretion <150 mg/day**.
- **Proteinuria >150 mg/day** means either **overproduction of non-albumin protein** (paraproteins in multiple myeloma) or **damage to the barrier or tubules**.
- **Albuminuria is a sensitive marker of glomerular injury**, because albumin excretion is controlled chiefly at the filtration barrier.
- **24-hour albumin excretion** needs concurrent creatinine to confirm completeness: expect **urine creatinine 20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.
- **Test hierarchy — Kidney Disease: Improving Global Outcomes (KDIGO) 2023:** spot **urine albumin-to-creatinine ratio (ACR)**, preferably early-morning — **first line** → spot **protein-to-creatinine ratio (PCR)** if ACR unavailable or large non-albumin proteinuria suspected (Bence Jones proteinuria) → **reagent strip** (automated or manual reading) **third line**, confirmed by formal testing.
- **Why ACR over PCR:** albuminuria is more sensitive to glomerular injury and to risk of adverse outcomes, and the assay is easier to standardise.
- **Trap:** the **reagent strip detects albumin only** and misses non-albumin protein such as light chains.
- **Prognosis:** large prospective cohorts establish rising albuminuria as an **independent risk factor for CKD progression, cardiovascular disease, stroke and death** — hence KDIGO albuminuria categories (see Chronic Kidney Disease).

Measurement of Kidney Function in a Dynamic State

- **eGFR equations are invalid when function is changing** — an anuric critically ill patient in the first 12 hours has a GFR of zero, whatever the creatinine-estimated GFR reads.
- **Guidelines therefore use change in SCr and urine output together** to identify acute kidney injury (AKI); KDIGO currently incorporates only these two.
- **Both are insensitive to early injury.** Novel biomarkers — urine **neutrophil gelatinase-associated lipocalin (NGAL)**, urine cystatin C, **kidney injury molecule-1 (KIM-1)** — may detect early, subclinical injury and predict recovery, pending further study.
- **Kidney injury continuum (Fig. 90.3):** normal → increased risk → damage → falling GFR → kidney failure → death, with complications arising at each step. Biomarkers identify susceptibility and injury mechanisms

during the **window for early targeted intervention**; creatinine and oliguria diagnose only once GFR has fallen (staging); biomarkers may later track progression. **GFR decline is a late, insensitive marker.**

Concept — Every kidney-function number rests on an assumption: creatinine assumes stable muscle mass and secretion, eGFR assumes a steady state, and the dipstick assumes the protein is albumin; identifying which assumption is broken tells you when the number is misleading.

ACUTE KIDNEY INJURY

- **Circulation of the nephron:** afferent arteriole → glomerular capillary bed → filtration across endothelium, GBM and podocytes into Bowman's space → efferent arteriole → peritubular capillaries alongside the tubules → venous system.
- **GFR is controlled by the Starling forces across the barrier** — the balance of hydrostatic and oncotic pressures favouring or opposing filtration. In health the drive is overwhelmingly toward filtration, **dominated by intracapillary hydrostatic pressure.**
- **Net filtration pressure (Fig. 90.4A):** 10 mm Hg = glomerular hydrostatic 60 – Bowman's capsule 18 – glomerular oncotic 32 mm Hg.
- **Maintaining glomerular hydrostatic pressure is the foundation of GFR**, achieved by adjusting relative afferent and efferent tone; in AKI something interferes with this process.

FIG. 90.4B ARTERIOLAR TONE, RENAL BLOOD FLOW AND NET ULTRAFILTRATION PRESSURE

Arteriolar change	Renal blood flow	Net ultrafiltration pressure (GFR)
Afferent resistance increased (constriction)	Falls	Falls
Afferent resistance decreased (dilatation)	Rises	Rises
Efferent resistance increased (constriction)	Falls	Rises
Efferent resistance decreased (dilatation)	Rises	Falls

A from Hall JE, Guyton and Hall Textbook of Medical Physiology, 12th ed, 2011; B Gilbert et al., NKF Primer, 8th ed, 2022, Fig. 2.5.

- **Pattern:** afferent changes move flow and filtration pressure **together**; efferent changes move them **in opposite directions.**
- **Trap:** efferent constriction lowers renal blood flow yet **raises** GFR — so blocking it (angiotensin II blockade) drops GFR when perfusion is low.

Definition of Acute Kidney Injury

- **KDIGO defines AKI by change in SCr and/or urine output** — either criterion alone suffices; stage is set by the size of the SCr rise or the duration of oliguria. AKI can also be classified by cause (below).

FIG. 90.5A KDIGO STAGES OF ACUTE KIDNEY INJURY

Stage	Serum creatinine criterion	Urine output criterion
1	Rise ≥ 0.3 mg/dL or $\times 1.5$ baseline	< 0.5 mL/kg/h for 6 h
2	$\times 2$ baseline	< 0.5 mL/kg/h for 12 h
3	$\times 3$ baseline, or SCr ≥ 4 mg/dL with an acute rise ≥ 0.5 mg/dL	< 0.3 mL/kg/h for 24 h, or anuria for 12 h

FIG. 90.5B KDIGO FUNCTIONAL AND STRUCTURAL CRITERIA FOR KIDNEY CONDITIONS

Condition	Functional criteria	Structural criteria
AKI	KDIGO AKI criteria (Fig. 90.5A)	No criteria
Acute kidney diseases and disorders (AKD)	AKI; or GFR <60 for <3 months; or GFR fall ≥35% or SCr rise >50% for <3 months	Kidney damage for <3 months
CKD	GFR <60 for >3 months	Kidney damage for >3 months
No known kidney disease (NKD)	GFR ≥60, stable SCr	No kidney damage

Gilbert et al., NKF Primer, 8th ed, 2022, Fig. 31.1.

- **Pattern:** the three-month mark divides acute from chronic; AKD captures subacute loss that meets neither AKI nor CKD criteria.

Epidemiology and Impact of Acute Kidney Injury

- **AKI is common, costly and deadly.** A large meta-analysis using KDIGO criteria found AKI in a minority of community patients but in **one-fifth to one-third of hospitalised patients**; pooled mortality about **one in four, rising to about one in two when renal replacement therapy (RRT) is required**.
- **United States burden:** over a million cases a year; several extra hospital days per episode; costs above 9 billion dollars annually; nearly 300,000 deaths — **AKI raises mortality more than diabetes, prostate cancer, breast cancer and heart failure combined**.
- **Even early-stage AKI confers excess risk of death**, so understanding its mechanisms is needed to prevent progression to RRT and death.

Overview of Acute Kidney Injury Aetiology, Diagnosis, and Management

- **Three categories: prerenal, intrinsic, postrenal** (Fig. 90.6), grouped by compartment; acute tubular necrosis (ATN) is the main intrinsic type.

FIG. 90.6 CAUSES OF ACUTE KIDNEY INJURY BY COMPARTMENT

Category	Compartment	Representative causes
Prerenal	Systemic perfusion	Hypovolaemia, cardiac failure, hepatorenal syndrome
Intrinsic	Renal artery	Renal artery occlusion; large- or medium-vessel vasculitis
Intrinsic	Renal vein	Renal vein thrombosis
Intrinsic	Small vessels	Thrombotic microangiopathy, renal atheroembolism, small-vessel vasculitis
Intrinsic	Glomerulus	Anti-GBM disease, lupus nephritis, postinfectious glomerulonephritis, infective endocarditis, membranoproliferative glomerulonephritis, cryoglobulinaemia, IgA nephropathy / Henoch–Schönlein purpura
Intrinsic	Tubule (ATN)	Ischaemia, nephrotoxins, rhabdomyolysis, radiocontrast
Intrinsic	Interstitium (acute interstitial nephritis, AIN)	Drugs, infections, systemic disease
Intrinsic	Intratubular obstruction	Cast nephropathy, drugs, crystalluria
Postrenal	Urinary outflow	Bladder outlet obstruction, tumours, calculi, papillary necrosis, retroperitoneal fibrosis

Modified from Floege J, Johnson RJ, Feehally J, *Comprehensive Clinical Nephrology*, 4th ed, 2010, Fig. 66.1.

KEY POINTS: ACUTE KIDNEY INJURY

- AKI may be multifactorial, with several events conspiring to cause injury.

- Speed of diagnosis matters: it allows insults to be reversed and further injury prevented.
- Prerenal AKI and ATN together cause nearly three-quarters of in-hospital AKI.

Prerenal Acute Kidney Injury

Illustrative case. Gastroenteritis in a hypertensive woman on lisinopril: hypotension, tachycardia, dry mucosa; SCr doubled, raised blood urea nitrogen (BUN); specific gravity 1.020, bland sediment, **fractional excretion of sodium (FeNa) 0.5%**. With resuscitation **SCr fell to near baseline within 2 days (Fig. 90.7).**

KEY POINTS: PRERENAL ACUTE KIDNEY INJURY

- Kidney structure is normal, so the urinalysis is bland — no active sediment.
- Correcting the prerenal insult produces immediate improvement.
- Drugs or conditions that impair autoregulation magnify the effect of a prerenal insult.
- FeNa is low because the kidney is reclaiming sodium to defend volume.

Aetiology and Mechanisms

- **Mechanism:** volume depletion → reduced renal perfusion → fall in glomerular capillary hydrostatic pressure → fall in GFR.
- **Autoregulation holds renal blood flow and GFR almost constant across a wide range of mean arterial pressure (Fig. 90.8), by dilating the afferent and constricting the efferent arteriole** — maximising inflow, restricting outflow.
- **Juxtaglomerular apparatus** = macula densa + glomerular tuft with its afferent and efferent arterioles (Fig. 90.2).

THE TWO AUTOREGULATORY MECHANISMS

	Myogenic reflex	Tubuloglomerular feedback (TGF)
Sensor	Afferent arteriolar smooth muscle responding directly to perfusion pressure	Macula densa (distal loop of Henle, abutting the tuft) sensing tubular NaCl
Response to low perfusion	Afferent dilatation, prostaglandin-mediated	Low NaCl delivery → renin from juxtaglomerular cells → angiotensin II → efferent constriction ; angiotensin II also raises proximal, and via aldosterone distal, sodium reabsorption
Response to high pressure	Afferent constriction protects the glomerulus	—
Disabled by	Non-steroidal anti-inflammatory drugs (NSAIDs) — block prostaglandins	RAAS blockade — abolishes efferent constriction

• Factors that magnify hypoperfusion by impairing autoregulation:

- **Structural (nonmodifiable)** — older age, CKD, kidney transplantation, hypertension, vascular disease alter renal vasculature so it responds poorly.
- **NSAIDs** — prevent prostaglandin-mediated afferent dilatation.
- **Sepsis and hepatorenal syndrome** — local renal vasoconstriction opposing efforts to maximise perfusion.
- **Hypercalcaemia and radiocontrast** — afferent vasoconstriction.
- **RAAS blockade** (as in the case) — removes the efferent constriction needed to hold glomerular pressure during ischaemia.

TABLE 90.3 NONMODIFIABLE AND MODIFIABLE RISK FACTORS FOR ACUTE KIDNEY INJURY

Nonmodifiable	Modifiable
CKD, diabetes mellitus, older age, chronic liver disease, congestive heart failure	Hypertension, hypoalbuminaemia, anaemia, sepsis, nephrotoxic drugs
Renal artery stenosis, peripheral vascular disease, acquired immunodeficiency syndrome (AIDS), prior kidney surgery	Mechanical ventilation, hypercholesterolaemia, rhabdomyolysis, hypercalcaemia

Gilbert et al., NKF Primer, 8th ed, 2022, Table 31.4.

- **Use:** nonmodifiable factors identify who needs protection; modifiable ones are what the surgeon can change before and after operation (drugs, sepsis, anaemia, ventilation, calcium).

Diagnosis

- **Weigh history, examination and laboratory data together** to build a case for or against each type; real patients rarely show every typical finding.
- **History:** large-volume losses; consider prerenal AKI even in a **normotensive patient with decompensated heart failure, sepsis or cirrhosis** (impaired autoregulation, reduced perfusion). After surgery, **review intraoperative and postoperative haemodynamics** for transient hypotension.
- **Examination:** tachycardia and hypotension — may be absent in low-flow states such as oedematous heart failure, where history corroborates.
- **Laboratory logic:** normal tubules conserve sodium (**>99% of filtered sodium can be reclaimed**) and water, and urea is passively reabsorbed.
- **Urine sodium is unreliable when urine output is low** → use FeNa; FeNa is misleading with **new diuretic therapy or metabolic alkalosis** → use **fractional excretion of urea (FeUrea)**. Both need simultaneous serum and urine sodium or urea and creatinine; calculators are widely available.
- **BUN:creatinine ratio >20:1** reflects passive urea reabsorption but is **not a reliable discriminator**, as non-AKI factors alter it.
- **High vasopressin (antidiuretic hormone, ADH)** concentrates the urine; the **response to correction is the single most important test**.

PRERENAL ACUTE KIDNEY INJURY VERSUS ACUTE TUBULAR NECROSIS

Index	Prerenal AKI	ATN
Urine sodium	<20 mmol/L	Not low, typically >20 mmol/L
FeNa	<1%	>1%
FeUrea (on diuretics or in metabolic alkalosis)	<35%	>35%
Specific gravity	>1.015 (concentrated)	<1.010 (dilute)
Sediment	Bland: no significant proteinuria, haematuria or granular, red cell or white cell casts; red cells, white cells or hyaline casts do not exclude it	Sloughed tubular epithelial cells; muddy-brown granular casts (may be absent early)
Response to removing insult	Immediate recovery	Lag: SCr rise slows, plateaus, then recovers over days to weeks
Biopsy	Rarely needed; normal parenchyma	Rarely needed; tubular injury with glomeruli and interstitium intact

- **Pattern:** every prerenal index is a sign of **working tubules defending volume**; every ATN index is a sign of **tubules that can no longer conserve sodium or water**.
- **Trap:** a high FeNa on a patient newly given diuretics does not prove ATN — use FeUrea.

Management

- **Timely reversal of the prerenal state, tailored to cause:** hypovolaemia → **isotonic fluid**; sepsis → treat and give volume as needed; heart failure → **diuretics and fluid restriction** to improve forward flow and decongest the kidneys; surgical patient with intraoperative hypotension → hold antihypertensives.
- **Uncorrected prerenal AKI evolves into intrinsic injury**; avoiding hypotension and further insults is paramount.
- **Hypotension is relative to the patient's baseline:** a mean arterial pressure (MAP) >65 mm Hg maintains GFR in healthy adults, but the **autoregulatory curve shifts right in older patients with vascular disease**, so a "normal" MAP may be too low.

FIG. 90.9 COMPOSITION OF INTRAVENOUS CRYSTALLOID SOLUTIONS

Fluid	Osmolarity (mOsm/L)	Effective osmolarity	Na	Cl	K	Buffer	Glucose (g/dL)
Human plasma	308	308	135–144	95–105	3.5–5.3	23–30	0.07–0.11
5% dextrose (D5)	252	0	0	0	0	0	5
D5 0.45% saline	406	154	77	77	0	0	5
D5 0.9% saline	560	308	154	154	0	0	5
0.45% saline	154	154	77	77	0	0	0
0.9% saline	308	308	154	154	0	0	0
Lactated Ringer's	273	273	130	109	4	28	0
Plasmalyte-148	294	294	140	98	5	50	0

Electrolytes in mmol/L; rows run from hypotonic to isotonic. Data from Moritz ML, Ayus JC, *N Engl J Med* 2015.

- **Dextrose is not an effective osmole** — it is metabolised quickly, so a D5 solution behaves as its saline base; D5 alone is effectively free water.
- **Trap:** lactated Ringer's and Plasmalyte are slightly hypotonic to plasma; only 0.9% saline (and D5 0.9% saline) match plasma effective osmolarity.

Intrinsic Acute Kidney Injury

- **Injury to the parenchyma — tubules, interstitium or glomeruli**, often more than one compartment.
- **Unlike prerenal AKI, parenchymal injury takes time to recover.**

FIG. 90.12 INTRINSIC ACUTE KIDNEY INJURY CATEGORIES

	Acute tubular necrosis	Acute glomerulonephritis	Acute interstitial nephritis
Cause	Ischaemic or nephrotoxic agents	Immune-complex, cell-mediated or anti-GBM	Allergic or non-allergic
Clinical pearls	Tubulotoxin exposure or prolonged prerenal state (sepsis, burns, gastrointestinal losses)	Palpable purpura, recent infection, lupus or pulmonary-renal syndrome	Recent infection, drug exposure (antibiotic, NSAID, checkpoint inhibitor) or systemic disease
Pathology	Tubular damage	Diffuse endocapillary or mesangial proliferation	Interstitial infiltrates
FeNa	>1%	>1% but may vary	>1% but may vary
Urine	Sloughed tubular cells; granular (muddy-brown) casts	Low-grade proteinuria; dysmorphic red cells, red cell casts	Urine eosinophils ± low-grade proteinuria; white cells, white cell casts

Images: Gilbert et al., *NKF Primer*, 8th ed, 2022, Figs. 4.2b, 4.2d and 33.1.

- **Pattern: the cast identifies the compartment** — granular for tubule, red cell for glomerulus, white cell for interstitium.

Acute Tubular Necrosis

Illustrative case. A vasculopath with chronic obstructive pulmonary disease (COPD) is hypotensive during partial nephrectomy, then needs ventilation and vasopressors with minimal urine output: **FeNa 3%, specific gravity 1.003, many granular casts**. Haemodynamics recover in 2 days, yet **SCr rises for about a week, plateaus, then falls (Fig. 90.10)**.

- **ATN is a misnomer:** injury extends beyond the tubules, and cell death is **more often apoptosis than necrosis**.
- **Two mechanisms: ischaemic and toxic.**

KEY POINTS: ISCHAEMIC ACUTE TUBULAR NECROSIS

- Hospital ATN is often a combination of ischaemic and toxic insults.
- Whether ischaemic or toxic, the mechanisms that lower GFR are similar.
- The tubules are the most vulnerable site for hypoxic injury.
- Diagnosis is supported by a lag in improvement after insults are removed, muddy-brown granular casts, and high urine sodium or FeNa.
- Recovery is possible but may take days to weeks.

Ischaemic acute tubular necrosis.

- **Cause:** a prolonged or extreme prerenal state.
- **Why tubules: they work at the brink of hypoxia even normally.** The kidney receives about 25% of cardiac output, but **medullary flow is progressively lower than cortical** because supply changes to countercurrent flow in the vasa recta.
- **Supply–demand mismatch (Fig. 90.11):** tissue oxygen tension falls from **50–100 mm Hg in cortex to 10–20 mm Hg in medulla** (osmolality 300 → 1200 mOsm), while the heavy-workload segments — **S3 segment of the proximal tubule and medullary thick ascending limb** — have high oxygen consumption; the outer medulla extracts a larger fraction of its oxygen supply than the heart, whereas the kidney as a whole extracts little.
- **GFR fall is multifactorial** — haemodynamic change, endothelial damage and inflammation.
- **Polarity sequence:** healthy cells are tightly packed and polarised (apical brush border; basolateral Na^+/K^+ -ATPase) → prolonged ischaemia **displaces Na^+/K^+ -ATPase to the cytoplasm or apical membrane** → proximal sodium reabsorption fails → **high sodium delivery to the macula densa activates TGF** → afferent vasoconstriction and RAAS shutdown → GFR falls.
- **This apparently maladaptive GFR fall is protective:** it cuts tubular workload and solute loss, giving time for regeneration.
- **If ischaemia persists:** cell death (necrosis or, more commonly, apoptosis) → sloughing into the lumen → **intratubular obstruction** and the **pathognomonic muddy-brown granular casts** — **dead tubular cells complexed with Tamm–Horsfall protein**.
- **Autoregulation is itself impaired in ATN** (partly via endothelial injury), further lowering GFR and heightening vulnerability to new insults; **peritubular capillary endothelial injury worsens outer medullary hypoxia**; hypoxia-induced inflammation amplifies every step.
- **Recovery** (regeneration → restored polarity → full basement membrane coverage) takes **days, weeks or months**; surviving cells are thought to dedifferentiate and proliferate to repopulate the tubule.

Toxic acute tubular necrosis.

Illustrative case. An obese man undergoes robotic nephroureterectomy **in flexed lateral decubitus for over 5 hours**; SCr and urea more than double and **creatinine phosphokinase (CPK) peaks above 21,000 IU/L at 24**

hours (normal 0–170). SCr improves slowly on 3–4 L/day lactated Ringer's. **He never had limb pain** — positional rhabdomyolysis can be silent.

KEY POINTS: TOXIC ACUTE TUBULAR NECROSIS

- Toxins injure through mechanisms that are not mutually exclusive: prerenal vasoconstriction causing ischaemic tubular injury, direct tubular injury and intratubular obstruction.
- Patient factors increase susceptibility to a given toxin — for example diabetes and contrast-induced injury.

- **Same end pathways as ischaemic ATN**; the trigger is an endogenous or exogenous toxin. **Most hospital ATN is mixed ischaemic and toxic.**

Postcontrast acute kidney injury.

- **Definition:** SCr rise typically **within 48 hours of intravenous iodinated contrast** — usually computed tomography (CT) in urology. Also called contrast nephropathy.
- **Mechanisms:** prerenal vasoconstriction (→ ischaemic ATN if prolonged or combined with other insults) and **direct tubulotoxicity**.
- **Risk factors:** older age, diabetes, CKD, anaemia, contrast **volume** and **route (arterial versus venous)**.
- **Prevention:** intravenous fluid before contrast in hospitalised patients is the most effective measure; optimal fluid and rate are unclear.
- **Dialysis to remove contrast does not reduce risk**, and dialysis patients given iodinated contrast **do not need urgent dialysis**.
- **Gadolinium-based magnetic resonance imaging (MRI) agents are not nephrotoxic at usual doses**, but **groups I and III are linked to nephrogenic systemic fibrosis (NSF)** — a progressive sclerosing disorder of skin and internal organs with high mortality. Avoid them in **AKI, CKD with eGFR <30 mL/min, and dialysis**. **Group II agents have little or no association with NSF** and may be considered in advanced kidney disease.

TABLE 90.4 GROUPING OF GADOLINIUM-BASED MRI CONTRAST AGENTS BY RISK OF NEPHROGENIC SYSTEMIC FIBROSIS

Group	Definition	Agents	Advanced kidney disease
I	Most NSF cases; older renal restrictions apply; no longer advertised in the United States	Gadodiamide (Omniscan), gadoversetamide (OptiMARK), gadopentetate dimeglumine (Magnevist)	Avoid
II	Few, if any, confirmed NSF cases	Gadobenate dimeglumine (MultiHance), gadoteridol (ProHance), gadobutrol (Gadovist), gadoterate meglumine (Dotarem)	May be considered
III	Insufficient data to classify; few, if any, cases	Gadoxetate disodium (Eovist)	Treated as group I

Gilbert et al., NKF Primer, 8th ed, 2022, Table 6.1.

- **Trap:** group III is not an intermediate-safety group — **it is handled as group I out of caution**; only group II is acceptable in advanced disease.

Myoglobin-induced injury.

- **Mechanism:** prolonged pressure in lateral decubitus → muscle necrosis → myoglobin release → three routes: **direct tubulotoxicity, intratubular precipitation and obstruction**, and **renal hypoperfusion** from fluid sequestration and inflammatory vasoconstriction → ischaemic tubular injury.

- **Urinary alkalinisation increases myoglobin solubility but has no proven benefit** and risks electrolyte harm.
- **Treatment: prevent further muscle injury and give early aggressive isotonic volume resuscitation.**

Diagnosis of acute tubular necrosis.

- **History of a prerenal or toxic insult** leads — review all medications and exposures; urinary indices and course as in the comparison table above.
- **Granular casts may be absent** early or missed through operator error.
- **Biopsy only if recovery lags** despite removed insults and clinical improvement, or other AKI types are suggested (Fig. 90.12).

Management of acute tubular necrosis.

- **Remove prerenal and toxic insults and prevent further ones** — "avoid hypotension and nephrotoxic agents"; this is the heart of recovery, since autoregulation is impaired in every AKI.
- **Avoid hypotension relative to baseline**; hold antihypertensives if needed; **stop RAAS blockers and NSAIDs** unless essential.
- **Dialysis** for life-threatening electrolyte disturbance, volume overload or uraemia not controlled medically (Table 90.5); it may be temporary or permanent, depending on severity, persistence and comorbidity; dialysis for weeks to months before recovery is rare.
- **Diuretics have one role only: symptomatic volume overload from reduced urine output** — they do not treat ATN.

TABLE 90.5 INDICATIONS FOR DIALYSIS USING THE MNEMONIC AEIOU

Indication	Key features
Acidosis	Metabolic acidosis unresponsive to bicarbonate, if appropriate
Electrolyte abnormalities	Most commonly hyperkalaemia
Ingestions	Most water-soluble compounds are dialysable; salicylates, alcohols and lithium commonly need dialysis
Overload	Volume overload causing respiratory distress unresponsive to other means of fluid removal
Uraemia	Subjective: malaise, altered (often metallic) taste, poor appetite, hypersomnia, tremor, altered mental state

- **Pattern:** each indication is **refractory to medical therapy** — AEIOU identifies failures of medical management, not biochemical thresholds.

Acute Interstitial Nephritis (AIN)

Illustrative case. Ten days after open urethroplasty a young man has nausea, **metallic taste** and anorexia; SCr risen fivefold; urine shows **many white cells, occasional eosinophils, trace protein**, sodium 50 mmol/L; echogenic kidneys without hydronephrosis. Biopsy: AIN; a **3-week prednisone taper** restores baseline SCr.

KEY POINTS: ACUTE INTERSTITIAL NEPHRITIS

- Infection and systemic disease can cause AIN, but drugs — particularly antibiotics and NSAIDs — are the commonest cause at all ages.
- AIN belongs in the differential diagnosis of sterile pyuria.
- Timely identification and removal of the offending agent is the key to treatment.

Aetiology and mechanisms.

- **Acute infiltration of the interstitium by inflammatory cells, with oedema**; the interstitium is normally a tiny compartment because tubules are packed closely.
- **Hypersensitivity reaction to an antigen — usually a drug or pathogen**. GFR falls through **mainly cell-mediated** (also antibody-mediated) injury → tubular damage, vascular injury and possibly obstruction from interstitial oedema.
- **Drugs cause the large majority**; infection a smaller share (Table 90.6).
- **Infective AIN**: consider in complicated urinary tract infection with *Escherichia coli*, *Campylobacter* or streptococci plus acute SCr rise.
- **Systemic causes (rare)**: sarcoidosis, systemic lupus erythematosus (SLE).
- **Timing**: classically a few days to weeks after starting the drug; rarely later in therapy.
- **Biopsy (Fig. 90.13)**: diffuse interstitial infiltrate, severe tubular injury with **tubulitis** (lymphocytes crossing the tubular basement membrane), and eosinophils.

TABLE 90.6 COMMON DRUGS ASSOCIATED WITH ACUTE INTERSTITIAL NEPHRITIS

Drug class	Examples (commonest in bold)
Antibiotics	β-lactams, sulfonamides, fluoroquinolones , rifampicin, vancomycin, erythromycin, ethambutol, chloramphenicol
Antivirals	Aciclovir, atazanavir, abacavir, indinavir
Analgesics	NSAIDs, selective cyclo-oxygenase-2 (COX-2) inhibitors
Gastrointestinal	Proton pump inhibitors (PPIs), H ₂ -receptor blockers, 5-aminosalicylates
Anticonvulsants	Phenytoin, carbamazepine, phenobarbital
Diuretics	Hydrochlorothiazide , furosemide, triamterene, chlorthalidone
Anticancer	Tyrosine kinase inhibitors, immune checkpoint inhibitors, B-Raf inhibitors
Other	Allopurinol, Chinese herbs

Gilbert et al., NKF Primer, 8th ed, 2022, Table 33.2.

- **Urological relevance**: the commonest culprits are the drugs urologists prescribe daily — penicillins, cephalosporins, fluoroquinolones, ibuprofen, ketorolac — and checkpoint inhibitors in uro-oncology.

Diagnosis.

- **Consider AIN in any unexplained hospital AKI**. Review outpatient drugs and the inpatient medication administration record; the **temporal link between exposure and AKI is critical**.
- **Extrarenal hypersensitivity** (low-grade fever, maculopapular rash, arthralgia, peripheral eosinophilia, transaminitis) may occur but is usually absent.
- **AIN is the commonest biopsy lesion in AKI on immune checkpoint inhibitors** (atezolizumab, pembrolizumab, nivolumab).
- **No single test is diagnostic**. Peripheral eosinophilia in context raises suspicion; urinalysis may be bland but **usually shows white cells — AIN is in the differential of sterile pyuria**; white cell casts or red cells are less common.
- **Urine eosinophils (Hansel stain) — the board-examination association is misleading**: they are insensitive in biopsy-proven AIN and also found in other diagnoses.
- **Proteinuria usually <1 g/day**; urine sodium and FeNa usually not low.
- **Nephrology consultation** to decide biopsy versus empirical treatment.

Management.

- **Mainstay: stop the offending drug or treat the offending pathogen**.

- **Early corticosteroids** improved renal recovery in observational studies (no randomised trials), suppressing T-cell inflammation and perhaps fibrosis; **oral and intravenous pulse steroids are equally effective** in prospective studies.
- **Prognosis:** only about half of antibiotic-induced AIN returns to baseline; **NSAID-associated AIN carries a poor prognosis.**
- **Systemic or infective AIN:** treat the underlying process.

Acute Glomerulonephritis

Illustrative case. An elderly never-smoker referred for **microhaematuria** has a **palpable purpuric rash**, hypertension, **SCr doubled in 4 days** and **red cell casts** — admitted for **rapidly progressive glomerulonephritis**, not cystoscopy.

KEY POINTS: ACUTE GLOMERULONEPHRITIS

- Glomerulonephritis is in the differential diagnosis of microscopic haematuria.
- Clues to glomerular haematuria include an acute SCr rise and proteinuria.
- Acute glomerulonephritis needs early nephrology review and kidney biopsy for diagnosis.

- **Third injurable compartment:** the glomerulus and its microvasculature.
- **Nephrotic versus nephritic patterns:** nephrotic diseases (focal segmental glomerulosclerosis, minimal change disease, membranous nephropathy) can cause AKI but usually present **insidiously with proteinuria or a creeping SCr**; nephritic disease is the acute, haematuric pattern.
- **Urological point of contact: haematuria.** Background, diagnosis and management of acute glomerulonephritis are in the electronic version.

Postrenal Acute Kidney Injury

- **Covered under Urinary Tract Obstruction and Obstructive Uropathy** (below and electronic version).

Management of Complications of Acute Kidney Injury

- **Commonest life-threatening complications: hyperkalaemia, acidaemia, volume overload and uraemia,** arising while the cause is being treated; hyperkalaemia and acidaemia are covered under electrolyte and acid-base disorders.

Volume Overload

- **Urgent only when causing respiratory distress** (or wound-healing problems from oedema); oedema or effusion alone does not demand urgent action.
- **Medical management:** sodium and fluid restriction plus intravenous diuretics.
- **Diuretic rules in AKI:** higher doses are needed → escalate the dose if the previous one failed → redose according to the drug's half-life for further removal → **loop diuretics first line**, adding a thiazide (metolazone, intravenous chlorothiazide) or carbonic anhydrase inhibitor (acetazolamide) for sequential blockade of sodium reabsorption.
- **Continuous infusion versus scheduled boluses** give similar 24-hour net diuresis; choice rests on logistics — reassessment, redosing and the higher level of care an infusion requires.

TABLE 90.7 COMMON DIURETICS AND DOSING

Drug	Bioavailability oral / intravenous (%)	Equivalent dose oral / intravenous (mg)	Duration (h)	Dosing interval
Furosemide	50, variable / 100	40 / 20	6	Daily to 6-hourly
Bumetanide	80–90 / 100	1 / 1	6	Daily to 12-hourly
Torsemide	80–90 / 100	20 / 20	8–12	Daily to 12-hourly
Ethacrynic acid	100 / 100	50 / 50	6	Daily to 8-hourly

From Partin AW, Dmochowski RR, Louis R, editors, *Campbell Walsh Urology*, 12th ed, 2020, Chapter 87, Table 7.

- **Pattern:** only furosemide has poor, variable oral absorption, so its **oral dose is double the intravenous dose**; the others are dosed the same by either route.
- **Torsemide lasts longest** (8–12 hours); the rest about 6 hours.

Uraemia

- **Symptoms (subjective):** malaise, poor appetite, dysgeusia, tremor — often confounded by other causes in hospital.
- **Signs:** increasing somnolence, confusion, **asterixis**, myoclonus, coarse tremor; a **pericardial rub** from uraemic pericarditis is rare in AKI; prolonged uraemia → obtundation and aspiration.
- **No medical therapy reverses the neurocognitive effects.**
- **Uraemic platelet dysfunction and bleeding** → **desmopressin (1-deamino-8-D-arginine vasopressin, DDAVP) 0.3 µg/kg intravenously or intramuscularly**, at the time of bleeding or 30–60 minutes before a procedure, redosed 12-hourly; efficacy not tested in high-quality studies. **Tachyphylaxis after 3 doses** — avoid prolonged use; monitor for hyponatraemia, hypertension and thrombosis.

Renal replacement therapy.

- **Indication:** progressive uraemia, hyperkalaemia, acidaemia or volume overload **refractory to medical therapy** — to prevent death. Goal: clear electrolytes and toxins and remove fluid.
- **Commonest inpatient modality: conventional intermittent haemodialysis** through an urgently placed temporary **dual-lumen 10–12 French catheter in the jugular or femoral vein**; dialysate runs countercurrent to blood through the filter.

CLEARANCE MECHANISMS IN RENAL REPLACEMENT THERAPY

	Diffusion	Convection
Driving force	Concentration gradient across the filter	Transmembrane pressure
What moves	Electrolytes and toxins, in or out of blood	Isotonic fluid, with solute carried along by solvent drag
Result	Solute clearance	Volume removal plus additional solute clearance

- **Continuous RRT** for haemodynamic instability — clearance and volume removal spread over longer periods with greater stability.
- **Trial evidence: no benefit from starting dialysis early**, before life-threatening complications; **continuous modalities are not superior to intermittent dialysis** for renal recovery or mortality.

- **Outcomes of AKI:** complete recovery; partial recovery (dialysis-independent but reduced function); non-recovery (dialysis-dependent AKI). **Persistence for 90 days or more** redefines these as CKD or end-stage kidney disease (ESKD) respectively.

Concept — *Location decides behaviour: prerenal AKI is a pressure problem in a structurally normal kidney, so tubules still conserve sodium and function returns the moment perfusion does, whereas intrinsic AKI damages the machinery itself, so sodium leaks, urine is dilute and recovery lags correction. Every diagnostic index in AKI is ultimately a test of whether the tubule still works.*

CHRONIC KIDNEY DISEASE

Illustrative case. A diabetic, hypertensive man awaiting partial nephrectomy has stable **eGFR 43 mL/min/1.73 m²** and **ACR 1.1 g/g** (1100 mg/g) and fears dialysis: **stage G3bA3 — very high risk** of complications, progression and adverse outcomes.

Definition and Epidemiology

- **KDIGO: abnormal kidney function or structure persisting for at least 3 months.**
- **Function = GFR and urinary albumin excretion;** persistent abnormality in either qualifies.
- **Structure alone qualifies** — a congenitally atrophic kidney with preserved GFR and no albuminuria is, by definition, CKD.
- **Risk of complications, cardiovascular disease and death rises with worsening GFR and albuminuria** → classification by **cause, GFR category and albuminuria category** (Fig. 90.16).

FIG. 90.16 KDIGO GFR AND ALBUMINURIA CATEGORIES AND PROGNOSTIC RISK

GFR category (mL/min/1.73 m ²)	A1 normal-mildly increased: ACR <30 mg/g (<3 mg/mmol)	A2 moderately increased: 30–300 mg/g (3–30 mg/mmol)	A3 severely increased: >300 mg/g (>30 mg/mmol)
G1 normal or high, ≥90	Low (no CKD without other markers)	Moderate	High
G2 mildly decreased, 60–89	Low (no CKD without other markers)	Moderate	High
G3a mildly to moderately decreased, 45–59	Moderate	High	Very high
G3b moderately to severely decreased, 30–44	High	Very high	Very high
G4 severely decreased, 15–29	Very high	Very high	Very high
G5 kidney failure, <15	Very high	Very high	Very high

A3 includes nephrotic range. Most United States adults sit in G1–G2 A1. From KDIGO CKD Work Group, KDIGO 2012 Clinical Practice Guideline, *Kidney Int Suppl* 2013 (prevalence data: National Health and Nutrition Examination Survey, 1999–2006).

- **Pattern:** risk rises **synergistically** down and across the grid — albuminuria can move a patient two risk bands without any change in GFR.
- **Trap: G1 and G2 are CKD only with albuminuria or structural damage;** G3–G5 qualify on GFR alone, but albuminuria is still needed to classify risk.
- **Prevalence:** over 30 million people in the United States and over 700 million worldwide.
- **Commonest causes: prolonged or poorly controlled diabetes and hypertension,** in the United States and worldwide. Others: glomerular disease with liver, autoimmune, cardiovascular disease or infection; **nephrolithiasis; chronic obstruction;** genetic kidney disease.

TABLE 90.8 RISK FACTORS FOR DEVELOPMENT OR PROGRESSION OF CHRONIC KIDNEY DISEASE

Modifiable	Nonmodifiable
Albuminuria, hypertension, episodes of AKI , underlying cause of kidney disease	Elevated plasma soluble urokinase receptor (suPAR), <i>APOL1</i> alleles
Obesity, hyperlipidaemia, smoking, high-protein diet	Black or Native American race, male sex, older age
Metabolic acidosis, hyperphosphataemia, hyperuricaemia, hyperglycaemia	Family history of kidney disease, low birth weight

Gilbert *et al.*, *NKF Primer*, 8th ed, 2022, Table 50.2.

- **Surgical relevance:** each perioperative AKI episode is itself a modifiable risk factor for CKD progression.

Diagnosis

- **Creatinine-based eGFR plus albuminuria**, measured as described under Measurement of Kidney Function.
- **Renal imaging, usually ultrasound, in any suspected or unevaluated CKD** to assess structural damage.
- Targeted evaluation of cause, tailored to history, under nephrology direction.

Management of CKD Complications

Anaemia

- **Erythropoietin (EPO) production falls with advancing CKD, particularly below GFR 30 mL/min/1.73 m² → progressive normocytic, normochromic anaemia.**
- **Treatment sequence:** ensure iron repletion and exclude other causes → EPO or equivalent, subcutaneous or intravenous.
- **Target haemoglobin 10–11.5 g/dL** in non-dialysis CKD and ESKD (KDIGO CKD 2023); **randomised trials showed harm when anaemia was corrected above 13 g/dL.**

Bone and Mineral Metabolism

- **Secondary hyperparathyroidism (HPT)** develops progressively: **reduced phosphate excretion, reduced 1,25-vitamin D production and hypocalcaemia** drive a normal parathyroid gland to respond.
- **Biochemical phenotype:** low to low-normal calcium, high-normal to high phosphate, **raised intact parathyroid hormone (PTH).**
- **Unchecked → bone disease and vascular calcification**, contributing to excess cardiovascular risk and death.
- **Treatment reverses the drivers:** dietary phosphate restriction, **phosphate binders taken with meals** to chelate gut phosphate, and 1,25-vitamin D.

SECONDARY VERSUS TERTIARY HYPERPARATHYROIDISM IN CKD

	Secondary	Tertiary
Gland	Normal gland responding appropriately to CKD biochemistry	Autonomous hyperfunction after prolonged untreated secondary HPT
Calcium	Low to low-normal	Unchecked hypercalcaemia; PTH unresponsive to correction of calcium and phosphate
Treatment	Phosphate restriction, binders, 1,25-vitamin D	Calcimimetics (stimulate the parathyroid calcium-sensing receptor) → subtotal parathyroidectomy if refractory

Acidaemia

- **Causes:** increased non-volatile acid production, increased bicarbonate loss and impaired renal acid excretion.

- **Alkali therapy (sodium bicarbonate or sodium citrate) to keep serum bicarbonate ≥ 22 mEq/L (KDIGO).**
- **Randomised trials: correcting acidaemia slows CKD progression** and benefits bone health and nutrition.

Hyperkalaemia

- **Renal potassium clearance needs two things: sodium delivery to the distal nephron and aldosterone** to drive distal sodium–potassium exchange.
- Advanced CKD reduces distal sodium delivery; **RAAS blockade creates a low-aldosterone state.**
- **Dietary potassium restriction limits fresh fruit and vegetables** → preferred strategy is **potassium binders and diuretics**, preserving beneficial RAAS blockers and a healthy diet.

Hypertension

- **Hypertension is both a cause and a consequence of CKD**; worsening function impairs sodium handling and causes volume retention.
- **Regimens need scheduled diuretics** to address volume excess. Loop diuretics were traditional in late CKD, but **thiazide or thiazide-like diuretics, particularly chlorthalidone, are effective and well tolerated in advanced CKD.**

Sexual Dysfunction

- **Sexual dysfunction worsens as GFR declines. Up to half of men with CKD or ESKD** report reduced libido and erectile dysfunction, with less frequent intercourse and poorer quality of life — **independent of other chronic comorbidity.**
- **Women:** amenorrhoea and infertility as GFR falls; pregnancy rates markedly reduced in advanced CKD and ESKD.
- **Putative mechanisms:** peripheral vascular disease, autonomic dysfunction, peripheral neuropathy, **hypogonadism**, mood disorders, **hyperprolactinaemia**, medication effects.
- **Reversibility:** variable improvement with dialysis, **more reliable improvement after transplantation — but not normalisation.**
- **Phosphodiesterase type 5 (PDE5) inhibitors may be effective in men.**
- **Under-addressed by providers**, possibly from discomfort with the topic.

Progression of Chronic Kidney Disease and Cardiovascular Risk

- **First principle: manage the underlying disorder.** Foundations: **glycated haemoglobin (HbA1c) $<7.5\%$ and blood pressure $<130/80$ mm Hg.**
- **Blood pressure control matters more than agent choice**, except in **proteinuric disease (with or without diabetes)**, where **RAAS blockade slows progression and lowers cardiovascular risk.**
- **Finerenone** — a non-steroidal selective mineralocorticoid receptor antagonist — reduces CKD progression and cardiovascular events in CKD with type 2 diabetes.
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors** give reno- and cardioprotection and a mortality benefit in large randomised trials, **in diabetic and non-diabetic patients alike.**
- **Progression is non-linear:** long plateaus punctuated by rapid decline, accelerated by non-adherence, poor diabetes and blood pressure control, nephrotoxins, sepsis and **surgical complications.**
- **Risk calculators predict progression to dialysis after nephrectomy or partial nephrectomy**, using baseline kidney function — the answer to the fearful patient in the case.

Concept — *CKD is staged on two axes because they carry independent risk: GFR measures how much kidney remains, albuminuria how actively it is being damaged. Each complication — anaemia, bone disease, acidosis, hyperkalaemia, hypertension — is simply the loss of one of the homeostatic functions described in the overview.*

COMMON ELECTROLYTE AND ACID–BASE ABNORMALITIES

Sodium Disorders

Illustrative case. After 4 days of watery diarrhoea, drinking only water, a hypotensive, tachycardic woman is confused with **sodium 122 mmol/L**, potassium 3.0 and bicarbonate 20 mmol/L.

KEY POINTS: SODIUM DISORDERS

- Sodium disorders are largely disorders of water balance, not sodium balance.
- Low serum sodium reflects water excess relative to sodium; high serum sodium reflects water deficit relative to sodium.
- If onset was more than 48 hours ago or is unknown, correct very slowly, with frequent sodium checks and adjustment for rapid change, to prevent irreversible neurological injury.

• Five organising principles:

1. **Volume status and water balance are separate entities.**
2. **Volume status is a clinical diagnosis** — profuse diarrhoea, hypotension, tachycardia and orthostasis indicate hypovolaemia; dyspnoea, orthopnoea, jugular venous distension and oedema with cardiac dysfunction indicate hypervolaemia.
3. **Sodium is the currency of volume**, so urine sodium is an adjunct to clinical assessment — **low urine sodium suggests a low-volume state.**
4. **Sodium disorders are disorders of water balance.**
5. **Water balance is a biochemical diagnosis** — low sodium means water excess, high sodium means water deficit.

- **Sites:** most sodium and water is handled by the proximal tubule, then loop of Henle and distal convoluted tubule, but the **principal cells of the distal nephron are the key site of water regulation.**
- **Requirement 1 — concentrated medullary interstitium:** the **ascending limbs are the only segments that absorb sodium while impermeable to water**, building the osmotic gradient that later draws water out of the collecting duct (reabsorbed water = concentrated urine = high urine osmolality).
- **Requirement 2 — apical aquaporin 2 (AQP2):** distal water reabsorption is impossible unless AQP2 is translocated to the **apical membrane of principal cells**, under **ADH (vasopressin)** acting on the **basolateral vasopressin type 2 (V2) receptor.**
- **ADH (Fig. 90.17):** made in the hypothalamus, secreted from the **posterior pituitary**; released in response to **raised plasma osmolality, low mean arterial pressure or low blood volume** — the most potent stimuli are **volume depletion and hyperosmolality**; hypothalamic osmoreceptors also drive thirst; V2 action in the medullary and papillary collecting ducts upregulates AQP2 production and insertion.
- **Serum osmolality = $2 \times \text{Na} + \text{glucose}/18 + \text{BUN}/2.8$** (glucose and BUN in mg/dL) — **sodium is the dominant contributor.**
- **Feedback:** high sodium → thirst and ADH → water retained if available and the medulla is concentrated; hypo-osmolality → thirst and ADH suppressed → water diuresis.
- **Volume depletion overrides hypo-osmolality** — the case: diarrhoeal sodium and water loss → volume depletion → ADH and water avidity despite falling osmolality → solute-free water intake → hyponatraemia.

Water distribution.

- **Total body water (TBW) = $0.6 \times \text{weight (kg)}$:** intracellular two-thirds, extracellular one-third; extracellular = **interstitial three-quarters, intravascular one-quarter.**

- **Osmotic forces govern intracellular ↔ extracellular movement:** isotonic fluid does not enter cells; hypotonic fluid distributes two-thirds to cells.
- **Starling forces govern interstitial ↔ intravascular movement,** so whatever the tonicity, only one-quarter of the extracellular share stays in vessels. **1 L of free water leaves only about 83 mL intravascular.**

FIG. 90.18 DISTRIBUTION OF 1 L OF INTRAVENOUS FLUID BETWEEN BODY COMPARTMENTS

Fluid	Intracellular (mL)	Interstitial (mL)	Intravascular (mL)
5% dextrose in water (D5W)	666	250	83
0.45% saline; D5 0.45% saline	333	500	167
0.9% saline; D5 0.9% saline	0	750	250
Lactated Ringer's	76	693	232
Plasmalyte-148	30	728	242
5% albumin	0	0	1000

Assumes plasma osmolality 308 mOsm/L.

- **Pattern:** dextrose adds nothing to distribution (not an effective osmole); only colloid stays in the vessel.
- **Trap:** even 0.9% saline leaves only a quarter of each litre intravascular.

Hyponatraemia

- **First confirm hypo-osmolality** (measured serum osmolality) before using the algorithm (Fig. 90.19).
 - **Hyperosmolar hyponatraemia** → **another effective osmole**, typically **hyperglycaemia**: glucose not entering cells becomes effective, draws water out of cells and dilutes sodium — not true hyponatraemia; apply a glucose correction factor.
 - **Iso-osmolar (pseudohyponatraemia)**: extreme hyperlipidaemia or paraproteinaemia interfering with measurement — largely eliminated by newer methods.
- **Hospital hyponatraemia is most commonly hypo-osmolar**, classified by volume status, urine sodium and urine osmolality.

FIG. 90.19 DIAGNOSTIC APPROACH TO HYPO-OSMOLAR HYPONATRAEMIA

Volume status	Urine test	Causes
Hypovolaemic: hypotension, tachycardia, no oedema	Urine Na >20 mmol/L (renal loss)	Diuretic excess, mineralocorticoid deficiency, osmotic diuresis, cerebral salt wasting
Hypovolaemic	Urine Na <20 mmol/L (extrarenal loss)	Vomiting, diarrhoea
Euvolaemic: no oedema, blood pressure not low	Urine osmolality high (above serum)	Syndrome of inappropriate ADH (SIADH), glucocorticoid deficiency, hypothyroidism, stress, drugs
Euvolaemic	Urine osmolality low	Psychogenic polydipsia and low-solute intake
Hypervolaemic: oedema, variable blood pressure and heart rate	Urine Na >20 mmol/L	Acute or chronic kidney failure
Hypervolaemic	Urine Na <20 mmol/L	Nephrotic syndrome, cirrhosis, heart failure

Urine osmolality compared with serum osmolality. Modified from Johnson RJ, Floege J, Tonelli M, *Comprehensive Clinical Nephrology*, 7th ed, 2023.

- **Hypo- and hypervolaemic states share low effective circulating (intravascular) volume** → kidney reclaims sodium (low urine sodium) → ADH and thirst appropriately stimulated → concentrated urine plus drinking → hyponatraemia.
- **Euvolaemic with urine osmolality above serum** → **ADH present without a stimulus ("inappropriate")**: extreme pain or nausea, drugs, infection, central nervous system (CNS) disease, paraneoplastic. With no

culprit it is **SIADH** — a **diagnosis of exclusion** after CKD, thyroid disease, adrenal disease and diuretics are excluded.

- **Euvolaemic with low urine osmolality** → **ADH appropriately absent**; water intake has outrun excretion, which is **solute-limited**: maximum dilution about **100 mOsm/L**, so 1000 mOsm of daily solute permits up to 10 L water excretion, but 100 mOsm permits only 1 L. Terms: **psychogenic polydipsia**, **beer potomania**, **"tea and toast" diet**.
- **Treatment by cause**: hypovolaemia → isotonic fluid (the case); heart failure → diuresis to improve effective circulating volume; **SIADH** → **free-water restriction**, then with nephrology guidance vasopressin receptor antagonists, salt tablets with furosemide, urea, lithium or demeclocycline; polydipsia → fluid restriction.
- **Rate of correction**: acute onset (<48 hours) may be corrected rapidly; slower or unknown onset → **rise no more than 8–12 mEq/L in 24 hours**.
- **Why**: brain cells adapt to chronic hypo-osmolality; rapid correction draws water out of brain cells → **osmotic demyelination**.
- **Highest risk is the case scenario**: isotonic fluid corrects hypovolaemia → removes the ADH stimulus → brisk water diuresis → rapid sodium rise. Monitor closely; if overcorrecting, **give DDAVP (ADH replacement) and free water**.
- **Severe symptoms** (lethargy, confusion, obtundation, seizures) → **3% hypertonic saline to raise sodium 4–6 mEq/L over a few hours**, then slow correction.

Hypernatraemia

- **Water deficit relative to sodium**: lack of water access with renal or extrarenal loss of solute-poor fluid, or sodium gain.
- **Diabetes insipidus (DI) pair**: **central** = lack of ADH; **nephrogenic** = renal ADH resistance; both impair water reclamation → water loss and dilute urine.
- **Nephrogenic DI is mostly acquired**: drugs (**lithium**, amphotericin B), electrolyte disturbance (**hypercalcaemia**, **hypokalaemia**) and **post-obstruction**; genetic causes exist.
- **Three practical rules**: intact thirst plus access to water prevents hypernatraemia → so an **undiagnosed DI patient kept nil by mouth postoperatively can become markedly hypernatraemic** → in a volume-overloaded hypernatraemic patient, free water adds little to overload because it distributes mainly into cells (Fig. 90.18).
- **Treatment: free water**, oral or parenteral (free-water content of fluids, Fig. 90.9).
- **Onset >48 hours or unknown** → **lower sodium no faster than 0.5 mEq/L per hour**: brain cells have adapted to hyperosmolality, and a rapid fall causes **cerebral oedema**, irreversible neurological damage and possible death.
- Calculate the free-water deficit to estimate infusion rate, but **check sodium very frequently** and adjust to avoid overcorrection.
- **Governing contrast**: overcorrecting hyponatraemia shrinks adapted brain cells (**osmotic demyelination**); overcorrecting hypernatraemia swells them (**cerebral oedema**).

Potassium Disorders

- **About 98% of the 3500 mmol total body potassium is intracellular**; the remaining ~70 mmol sits in extracellular fluid at 4–5 mEq/L.
- **The intracellular-to-extracellular ratio sets the resting membrane potential**; disruption causes muscular and neuronal dysfunction — at the extremes **cardiac arrest (hyperkalaemia)** or **paralysis (hypokalaemia)**.

- **Two lines of defence:** shifts between cells and extracellular fluid (**early**) and renal and gut excretion (**slower**); so every potassium disorder is a **shift** or an **excretion–intake mismatch**.
- **Daily balance:** diet about 100 mmol; kidney excretes **90–95% over 6–12 hours**; gut 5–10%, a minor role in health.
- **Acute load: insulin and β_2 -adrenergic stimulation increase Na^+/K^+ -ATPase activity**, moving potassium into cells — the basis of emergency therapy.
- **Renal excretion is distal tubule and collecting duct secretion.** Secretion is increased by:
 1. **High urine flow** — more distal sodium delivery for exchange; low flow reduces excretion.
 2. **Greater distal sodium delivery and absorption**, which makes the lumen electronegative and favours potassium (and hydrogen) secretion — **loop and thiazide diuretics cause hypokalaemia**; volume depletion or **potassium-sparing diuretics** do the reverse.
 3. **Rising aldosterone** — increases distal sodium reabsorption and hence potassium excretion.
 4. **Rising plasma potassium** acting directly on apical secretory channels.
 5. **Extracellular pH** — **metabolic alkalosis enhances, metabolic acidosis impairs** secretion.
- **Adaptation:** chronic high intake increases the efficiency of all five mechanisms.

Illustrative case. A diabetic man with CKD on a **β -blocker, a recently increased angiotensin-converting enzyme (ACE) inhibitor and insulin** has potassium 6.7 mEq/L, glucose 397 mg/dL and **peaked T waves**; calcium gluconate, insulin and an intravenous loop diuretic restore 4.9 mEq/L — **several simultaneous hits**.

Hyperkalaemia

- **Rarely from intake alone** while regulation is intact; results from two non-exclusive mechanisms:
 - **Shift out of cells** — insulin deficiency, cell lysis, **systemic acidosis** (hydrogen enters cells in exchange for potassium).
 - **Impaired renal secretion** — acute or chronic kidney disease, aldosterone deficiency or resistance, reduced distal sodium delivery and urine flow (volume depletion).
- **Urological cause: ureteric diversion into jejunum** (much less often ileum) → **intestinal reabsorption of urinary potassium**.
- **Pseudohyperkalaemia** — falsely high value with normal true level: traumatic difficult draw, prolonged tourniquet, excessive fist clenching; extreme leucocytosis or thrombocytosis. Exclude it first.

TABLE 90.9 CAUSES OF HYPERKALAEMIA BY MECHANISM

Mechanism	Causes
Pseudohyperkalaemia	Haemolysis, thrombocytosis, severe leucocytosis, fist clenching
Decreased kidney excretion	Acute or chronic kidney disease; aldosterone deficiency (type IV renal tubular acidosis, diabetic nephropathy, chronic interstitial nephritis, obstructive uropathy , adrenal insufficiency); drugs inhibiting potassium excretion; diseases impairing distal tubule function (sickle cell disease, SLE)
Abnormal distribution	Insulin deficiency, β-blockers , acidaemia, familial hyperkalaemic periodic paralysis
Abnormal release from cells	Rhabdomyolysis, tumour lysis syndrome

Gilbert et al., *NKF Primer*, 8th ed, 2022, Table 10.3.

- **Management principles (sequence):**
 1. **Judge urgency** — acuity of rise, absolute level (**>6.0 mEq/L generally urgent**, allowing for baseline) and neuromuscular or cardiac effect. Weakness ascends from the legs to trunk and arms, potentially to paralysis with **respiration spared**.
 2. **Electrocardiogram (ECG; Fig. 90.21):** peaked T (earliest) → prolonged PR, wide QRS → P loss → sine wave. **Any ECG change → intravenous calcium** to stabilise the myocardium, buying time.

3. **Identify the cause and tailor therapy** (Table 90.9).

4. **Shift or remove potassium** (Fig. 90.20).

FIG. 90.20 ALGORITHM FOR MANAGEMENT OF HYPERKALAEMIA

Step	Emergent (ECG changes, or rapid rise as in rhabdomyolysis, tumour lysis, crush injury)	Non-emergent
First	Stabilise membrane: calcium gluconate 10 mL of 10% intravenously	Exclude pseudohyperkalaemia (especially without risk factors)
Shift	Insulin 10 U regular intravenously with 50 mL 50% dextrose; salbutamol (albuterol) 20 mg in 4 mL nebulised	Dietary potassium counselling; avoid salt substitutes
Remove	Oliguria or ESKD → haemodialysis ; hypervolaemic non-oliguric → diuretics; metabolic acidosis → sodium bicarbonate; then consider potassium binders	Stop interfering drugs; ask about over-the-counter NSAIDs and herbal preparations; effective diuretic therapy (loop if eGFR <30 mL/min); oral sodium bicarbonate for acidosis; potassium binders (allow RAAS inhibitor dosing)

Gilbert et al., NKF Primer, 8th ed, 2022.

GUT POTASSIUM BINDERS (FIG. 90.20)

	Sodium polystyrene sulfonate (SPS, Kayexalate)	Patiromer (Veltassa)	Sodium zirconium cyclosilicate (Lokelma)
Mechanism	Sodium–potassium exchange resin, often with sorbitol; non-selective (potassium, calcium, magnesium)	Calcium-based non-absorbed polymer; exchanges calcium for potassium; binds magnesium	Non-polymer; preferentially captures potassium in exchange for sodium and hydrogen
Onset	Hours to days, variable	About 7 h ; no bowel movement needed	About 1 h
Site	Colon	Colon	Entire intestinal tract
Adverse effects	Diarrhoea, metabolic alkalosis, hypernatraemia, volume overload, rarely colonic necrosis ; separate other oral drugs by ≥3 h	Constipation, diarrhoea, flatulence, hypomagnesaemia ; separate other oral drugs by ≥3 h	Constipation, diarrhoea, oedema ; raises gastric pH, affecting drugs with pH-dependent solubility
US Food and Drug Administration (FDA) approval for emergencies	Yes — the only one	No	No

- **Trap:** the only binder approved for emergencies is the worst one — SPS is modestly effective, slow and linked to **intestinal necrosis, especially postoperatively or with bowel disease or ileus**; despite low cost it is **not first-line**. Patiromer and sodium zirconium cyclosilicate are increasingly used acutely.

Hypokalaemia

- **Mechanism:** excessive shift into cells, or losses — renal, gastrointestinal or, rarely, sweat.

TABLE 90.10 CAUSES OF HYPOKALAEMIA BY MECHANISM

Mechanism	Causes
Inadequate intake	Severe malnutrition
Extrarenal losses	Vomiting, diarrhoea
Urinary losses	Non-potassium-sparing diuretics (loop, thiazide, acetazolamide); osmotic diuresis (hyperglycaemia); with hypertension : primary hyperaldosteronism, Liddle syndrome; with normal blood pressure : type 1 and type 2 renal tubular acidosis, Bartter syndrome, Gitelman syndrome, hypomagnesaemia
Abnormal distribution	Insulin administration, catecholamine excess (acute stress), thyrotoxic hypokalaemic paralysis, familial hypokalaemic periodic paralysis

Gilbert et al., NKF Primer, 8th ed, 2022, Table 10.3.

- **Often multifactorial** — e.g., refeeding a malnourished patient: low total stores plus an **insulin surge shifting potassium into cells**.
- **Rarely from intake alone**, because the kidney can greatly reduce excretion.
- **Lower gastrointestinal losses (diarrhoea) lower potassium directly** — intestinal and colonic potassium content is high.
- **Trap — vomiting**: gastric potassium content is minimal, so loss is not the cause; **acid loss → metabolic alkalosis → renal sodium bicarbonate excretion → increased distal sodium delivery → kaliuresis**.
- **Clinical**: neuromuscular instability and arrhythmia. **Severe acute hypokalaemia (<2.5 mEq/L)** → weakness in a pattern like hyperkalaemia, cramps, **rhabdomyolysis**. ECG: premature atrial and ventricular beats, sinus bradycardia, junctional rhythm, heart block, ultimately ventricular tachycardia or fibrillation; risk rises with severity, coexisting electrolyte disturbance and comorbidity.
- **Treatment — replacement**, with formulation and rate set by **ongoing losses** (profuse diarrhoea or aggressive diuresis needs more).
- **Potassium chloride is the commonest and most effective formulation**; potassium phosphate if hypophosphataemic, potassium citrate if acidotic.
- **Oral preferred**; intravenous only if oral impossible, because it raises potassium more slowly and needs significant volume. Usual dose **20–40 mEq, repeated** to normokalaemia.
- **Redistribution causes → replace cautiously**: correcting the cause can produce **rebound hyperkalaemia**. Symptomatic hypokalaemia needs urgent replacement regardless of cause.

FIG. 90.21 ELECTROCARDIOGRAPHIC CHANGES WITH POTASSIUM DISORDERS

Serum potassium (mEq/L)	Change
≈10	Ventricular fibrillation
≈9	Atrial standstill, intraventricular block
≈8	Prolonged PR interval, high T wave, depressed ST segment
≈7	High (peaked) T wave
3.5–5	Normal
≈3.0	Low T wave, high U wave (Q–T becomes Q–U)
≈2.5	Low T wave, high U wave, low ST segment

Gilbert et al., NKF Primer, 8th ed, 2022, Fig. 10.3.

- **Governing contrast**: hyperkalaemia **peaks the T wave and widens the QRS**; hypokalaemia **flattens the T wave and raises the U wave**.

Calcium Disorders

Illustrative case. Months of abdominal pain, polyuria, bone ache and low mood, then a first stone; dry and disoriented. **Calcium 12.9 mg/dL** (9.9 six months before), **phosphate falling to 2.7 mg/dL**, normal albumin.

- **Nearly 99% of body calcium is in bone**; <1% is plasma calcium, partly bound (albumin, organic anions) and partly **free ionised** — the **biologically active fraction** sensed by tissues.
- **Bone**: resorption raises serum calcium; formation lowers it.
- **Gut**: absorption is incomplete, reduced by calcium-binding anions, and involves passive and **calcitriol-dependent active** small-intestinal transport.
- **Kidney**: reabsorption proximal convoluted tubule (PCT) > thick ascending limb (TAL) > distal convoluted tubule (DCT).

RENAL CALCIUM HANDLING: COUPLED VERSUS UNCOUPLED SEGMENTS

	PCT and TAL	DCT
Link to sodium	Calcium follows sodium	Uncoupled
Volume depletion	Maximal sodium, hence calcium, reabsorption	—
Diuretic effect	Volume loading or loop diuretic → sodium dumping → calciuria	Thiazide blocks the Na–Cl cotransporter; calcium does not follow; resulting volume contraction increases proximal sodium and calcium reabsorption → reduced urinary calcium
Clinical use	Saline ± loop diuretic in hypercalcaemia	Thiazide for hypercalciuria in calcium stone formers

- **Calcitriol is made in the kidney** from 25-hydroxyvitamin D.
- **Direct regulators**: PTH, calcitriol, serum calcium itself and, to a lesser degree, calcitonin.
- **Response to hypocalcaemia (Fig. 90.22)**: PTH rises → bone turnover increases, releasing calcium and phosphate → kidney increases calcium reabsorption, reduces phosphate reabsorption and **increases 1 α -hydroxylase** → calcitriol increases intestinal calcium and phosphate absorption → calcium normalises with phosphate normal. The converse occurs in hypercalcaemia.
- **Humoral hypercalcaemia of malignancy: parathyroid hormone-related peptide (PTHrP)** from squamous, renal, bladder, breast and ovarian cancers mimics hyperparathyroidism.

DISCORDANT TOTAL AND IONISED CALCIUM

	Low total, normal ionised	Normal total, low ionised
Setting	Hypoalbuminaemia (40% of total calcium is albumin-bound; ionised calcium stays tightly regulated)	Acute respiratory alkalosis (hyperventilation)
Mechanism	Less bound calcium; free fraction preserved	Fewer hydrogen ions on albumin → more binding sites for calcium → ionised calcium falls
Action	Add 0.8 mg/dL to total calcium for every 1 g/dL albumin below 4 g/dL ; confirm with ionised calcium	Symptomatic hypocalcaemia occurs before compensation restores ionised levels

Hypercalcaemia

- **Urological context**: found during the initial metabolic work-up of a stone former. Most hypercalciuria is idiopathic, but **hypercalcaemia must be excluded** as its cause.

- **Diagnostic algorithm:**

1. **Intact PTH with a simultaneous serum calcium** — PTH cannot be interpreted without calcium. **Hypophosphataemia is a clue to PTH excess** (renal phosphate wasting). **Non-suppressed PTH with high calcium → primary hyperparathyroidism.**
2. **PTH suppressed → measure PTHrP** (humoral hypercalcaemia of malignancy) and **1,25-vitamin D** (extrarenal production in sarcoidosis or lymphoma).
3. **Neither raised →** increased bone resorption (prolonged immobilisation, metastases, hyperthyroidism) or reduced renal calcium excretion from drugs (**thiazides**).
4. **Rarely, hypercalcaemia with low urine calcium (<200 mg/g creatinine) → familial hypocalciuric hypercalcaemia (FHH)**, a genetic calcium-sensing disorder that is completely benign.

- **Features:** bone pain, muscle weakness, kidney stones, **polyuria from acquired nephrogenic DI**, constipation, nausea, depression; **severe hypercalcaemia may cause AKI**, partly via prerenal vasoconstriction.

- **Treatment is temporising until the cause is treated.** Indications: moderate-to-severe hypercalcaemia with symptoms, or **severe (≥ 14 mg/dL) even without symptoms.**

ACUTE TREATMENT OF HYPERCALCAEMIA

Agent	Mechanism	Onset, duration and use
Isotonic saline — mainstay	Exploits sodium–calcium coupling in PCT and loop	200 mL/h , then titrate to urine output 100–150 mL/h for 24–48 h , if no overload, heart or kidney failure; alone it is not definitive
Loop diuretic	Calciuresis at TAL	Only with fluid overload or heart or kidney failure , added to saline; not used otherwise
Calcitonin (intramuscular or intravenous)	Temporising	Rapid; tachyphylaxis within 48 h , little effect thereafter
Intravenous bisphosphonate	Inhibits bone resorption	Onset 2–4 days; effect lasts months ; sustained control
Denosumab	Monoclonal antibody reducing bone resorption	When bisphosphonates are contraindicated

- **Hypocalcaemia and phosphate disorders** are covered in the electronic version only.

Acid–Base Disorders

Illustrative case. A stone former with a new **mixed calcium oxalate–calcium phosphate stone**, recently **started on topiramate**: bicarbonate 21 mEq/L with a normal anion gap; 24-hour urine **pH 6.81**, **citrate low**, oxalate and sodium high, calcium phosphate supersaturation high — carbonic anhydrase inhibition (below).

- **Extracellular pH is held at 7.38–7.42** for life-sustaining processes.
- **Volatile acid (carbon dioxide)** from carbohydrate and fat metabolism is removed by ventilation; **non-volatile (fixed) acid** from carbohydrate, fat and sulphur- and phosphate-containing amino acids is handled by the kidney, which (1) **excretes the acid load** and (2) **reclaims filtered bicarbonate**.
- Disorders arise when ventilation, renal processes or both fail or are overwhelmed.

ACID–BASE TERMINOLOGY

Disorder	Primary change and threshold	pH	Common causes
Acidaemia / alkalaemia	pH <7.35 / pH >7.45	—	Respiratory, metabolic or combined
Respiratory acidosis	Hypoventilation; partial pressure of CO ₂ (pCO ₂) >45 mm Hg	Falls	Opioid respiratory depression, COPD air trapping
Respiratory alkalosis	Hyperventilation; pCO ₂ <35 mm Hg	Rises	Panic attacks, pregnancy, liver disease
Metabolic acidosis	Bicarbonate <22 mEq/L	Falls	Anion gap (AG): lactic acidosis, diabetic ketoacidosis; non-AG: renal tubular acidosis (RTA), diarrhoea
Metabolic alkalosis	Bicarbonate >29 mEq/L (net bicarbonate gain or acid loss the kidney cannot correct)	Rises	—

- **Bicarbonate source:** the chemistry panel value is **measured**; the blood gas value is **calculated** from pH and pCO₂ by the Henderson–Hasselbalch equation.
- **Anion gap = Na⁺ – (Cl[–] + HCO₃[–]); raised (>12)** by unmeasured organic acids. Albumin forms part of the normal gap, so **add 2.5 mEq/L for every 1 g/dL albumin below 4.0 g/dL**. AG and non-AG acidosis may coexist.
- **Compensation:** renal compensation for respiratory disorders takes 1–3 days; respiratory compensation for metabolic disorders is maximal by 24 hours; compensation never restores pH to normal, only toward it.
- **A low serum bicarbonate — the urologist's usual clue — has three explanations:**
 1. **Compensation for respiratory alkalosis** — proven only by blood gas: pH high. If metabolic acidosis is primary, pH is low.
 2. **AG metabolic acidosis** — the rise in AG should account for the fall in bicarbonate. Example: AG 18 from a baseline of about 6 (a rise of 12) predicts bicarbonate about 12 below the normal ~24 mEq/L. **Bicarbonate lower than predicted → an additional non-AG acidosis (mixed disorder).**
 3. **Non-AG metabolic acidosis** — renal or gastrointestinal.

Renal causes — proximal mechanism.

- **The proximal tubule reabsorbs 85–90% of filtered bicarbonate;** the rest is reclaimed distally.
- **Sequence:** hydrogen ion secreted into the lumen → combines with filtered bicarbonate → **luminal carbonic anhydrase** forms CO₂ and water → lipid-soluble CO₂ enters the cell → recombines with hydroxyl to form bicarbonate → exits basolaterally via the **Na–HCO₃ cotransporter** into blood.

TABLE 90.12 CAUSES OF PROXIMAL RENAL TUBULAR ACIDOSIS (TYPE II)

Defect	Causes
Isolated bicarbonate reabsorption defect	Carbonic anhydrase inhibitors , carbonic anhydrase deficiency
Generalised proximal transport defect (examples)	Cystinosis, multiple myeloma , light-chain disease , amyloidosis, vitamin D deficiency, ifosfamide , cidofovir, lead, aminoglycosides

Gilbert et al., NKF Primer, 8th ed, 2022, Table 13.5.

- **Urological presentation — topiramate and zonisamide** inhibit both proximal and distal carbonic anhydrase. Distal hydrogen secretion is intact, so **serum bicarbonate falls only mildly but urine pH is typically >6.0**.
- **Trap:** bicarbonate therapy exceeds the reabsorptive threshold, **raising urine pH further**, and the extra distal bicarbonate delivery causes **kaliuresis and hypokalaemia**.

Renal causes — distal mechanism.

- **Acid excretion = distal hydrogen secretion with urinary buffers: titratable acid** (chiefly phosphate, a largely fixed quantity) and **ammonia**, whose proximal generation is adjustable — **increased by metabolic acidosis and hypokalaemia, reduced by hyperkalaemia.**
- **Distal acidification defects:**
 1. **CKD** — fewer nephrons produce less ammonia, especially **below eGFR 40 mL/min/1.73 m²**; corrected with alkali (Table 90.13).
 2. **Hypoaldosteronism** (primary or secondary) → reduced potassium excretion → hyperkalaemia suppresses ammoniogenesis = **type IV RTA.**
 3. **Distal RTA type 1** — impaired distal hydrogen secretion (**H⁺-ATPase or Cl⁻/HCO₃⁻ exchanger**), acquired or congenital → low serum bicarbonate, **hypokalaemia** (potassium secreted in place of hydrogen), **high urine pH, hypocitraturia** (proximal citrate reabsorption rises with intracellular acidosis), ± **hypercalciuria** (bone buffering).
 4. **Distal RTA with hyperkalaemia** — a generalised distal defect in both hydrogen and potassium secretion (as in **urinary obstruction**), or a distal sodium transport defect that abolishes the voltage gradient favouring hydrogen and potassium secretion (Table 90.14).
- **Urological presentation — distal RTA type 1: a calcium stone former with low or borderline bicarbonate, high urine pH, mild hypokalaemia, hypocitraturia ± hypercalciuria** may have complete or incomplete (milder) distal RTA. Treat any underlying disorder and give **alkali** (Table 90.13).

RENAL TUBULAR ACIDOSIS COMPARED

	Proximal (type II)	Distal (type 1)	Type IV	Distal with hyperkalaemia
Defect	Bicarbonate reclamation (carbonic anhydrase, generalised proximal)	Distal H ⁺ secretion	Hypoaldosteronism → hyperkalaemia → low ammoniogenesis	H ⁺ and K ⁺ secretion, or distal Na transport voltage
Serum potassium	Low with bicarbonate therapy	Low	High	High
Urine pH	High on carbonic anhydrase inhibitors	High	—	—
Stone relevance	Topiramate, zonisamide	Hypocitraturia ± hypercalciuria → calcium stones	—	Obstruction

TABLE 90.13 COMMON ALKALI THERAPY REGIMENS

Preparation	Usual dose per unit	Comments
Sodium bicarbonate	650 mg = 8 mmol	May cause gastric gas
Sodium citrate/citric acid (liquid)	1 mmol sodium and citrate per mL	1 mmol citrate = 1 mmol bicarbonate ; avoid aluminium-containing drugs (antacids, sucralfate)
Potassium citrate (tablet)	5 and 10 mmol per tablet	Simultaneous potassium and alkali therapy
Citric acid/potassium citrate/sodium citrate (liquid)	1 mmol sodium and potassium and 2 mmol citrate per mL	Avoid aluminium-containing drugs
Potassium citrate (liquid)	2 mmol potassium and 2 mmol citrate per mL	Avoid aluminium-containing drugs

Modified from Johnson RJ, Floege J, Tonelli M, *Comprehensive Clinical Nephrology*, 7th ed, 2023.

- **Pattern:** citrate is a bicarbonate equivalent; **potassium citrate is the choice when hypokalaemia accompanies the acidosis** (distal RTA, carbonic anhydrase inhibitors); every citrate liquid interacts with aluminium.

TABLE 90.14 CAUSES OF DISTAL RENAL TUBULAR ACIDOSIS WITH HYPERKALAEMIA

Category	Causes
Disease	Lupus nephritis, obstructive nephropathy , sickle cell anaemia, voltage defects, familial causes
Drugs	Amiloride, triamterene, trimethoprim, pentamidine

Gilbert et al., *NKF Primer*, 8th ed, 2022, Table 13.6.

- **Pattern:** every listed drug **blocks the distal epithelial sodium channel**, removing the lumen-negative voltage that drives hydrogen and potassium secretion.

Gastrointestinal losses and the urine anion gap.

- **Extrarenal bicarbonate loss** exceeding the kidney's capacity to reclaim bicarbonate or acidify urine.
- **Urine AG = (urine Na⁺ + urine K⁺) – urine Cl[–]** — a surrogate for the ammoniagenic response to an acid load, because excreted ammonium carries chloride.
- **Negative urine AG = increased ammonium excretion = appropriate renal response → non-renal (gastrointestinal) cause.**
- **Positive urine AG = no rise in ammonium excretion → the kidney is responsible (RTA).**
- **Classic stone example — laxative abuse:** chronic diarrhoea causes large potassium, bicarbonate, sodium and water losses; potassium depletion stimulates ammonia production (urine AG negative); ammonia buffering can **raise urine pH (>5.8) despite alkali loss**; above the urate pKa (5.7) urate salt rather than uric acid predominates and combines with abundant ammonium → rare **ammonium acid urate stones**.

Concept — Every electrolyte disorder is read through two questions: is this a shift between compartments or a change in total body content, and has the kidney responded appropriately? Sodium is a water problem judged by ADH, potassium a distribution-then-secretion problem judged by distal sodium delivery and aldosterone, calcium a sodium-coupled problem judged by PTH, and non-AG acidosis a question of whether the kidney is making ammonium.

URINARY TRACT OBSTRUCTION AND OBSTRUCTIVE UROPATHY

Introduction

- **Urinary tract obstruction (UTO)**: blockage of urinary flow at any level from **renal tubule to urethra**, arising in fetal life, childhood or adulthood; usually shown as dilatation on imaging; congenital or acquired, benign or malignant.
- **Impact depends on degree** (partial or complete, unilateral or bilateral), **chronicity** (acute or chronic) and **pre-existing renal function**.

FIG. 90.23 HYDRONEPHROSIS, OBSTRUCTIVE UROPATHY AND OBSTRUCTIVE NEPHROPATHY

Term	Definition	Creatinine and GFR
Hydronephrosis	Dilatation of renal pelvis and/or calyces, with or without obstruction	No effect if unobstructed
Obstructive uropathy (OU)	Functional or anatomic obstruction of urinary flow at any level	GFR may or may not change, depending on the contralateral kidney
Obstructive nephropathy (ON)	Irreversible loss of renal function (permanent parenchymal injury)	GFR falls and creatinine rises, especially if bilateral

- **Asymptomatic OU may go unnoticed in SCr**, but uncorrected unilateral OU causes progressive irreversible ipsilateral loss, and **uncorrected bilateral OU leads to ESKD**.
- **OU is reversible if identified and corrected promptly** — one of the few potentially curable forms of kidney disease, so **consider it in any unexplained acute or chronic kidney disease**.
- Further detail: Chapter 51 and electronic version.

Postobstructive Diuresis

- **A normal physiological event after relief**, particularly of **prolonged and bilateral** obstruction; its size reflects the volume overload and the urea and electrolytes retained during obstruction.
- **Usually self-limited**; no defining rate, but **urine output up to 300 mL/h is common**. Polyuria typically begins within hours of relief in most patients and lasts about 2 days; fractional sodium excretion may be many times normal.
- **Three mechanisms**:
 1. **Transporter dysregulation** — sodium (Na^+/K^+ -ATPase) and water channels (all aquaporins, chiefly AQP2) are dysregulated during obstruction. In extreme cases **AQP2 suppression lasts days to weeks** → **nephrogenic DI**: polyuria, **low urine osmolality (usually 300–400 mOsm/kg water)** and, if severe, life-threatening hypernatraemia from free-water loss; diagnosed by water deprivation or desmopressin testing.
 2. **Extracellular volume expansion** (especially bilateral) → on relief, **reduced sympathetic tone, reduced aldosterone and increased atrial natriuretic peptide (ANP)** promote salt and water excretion.
 3. **Osmotic diuresis** — retained urea enters the filtrate, raising tubular osmotic pressure and reducing water reabsorption.
- **Commonest setting: catheterisation for urinary retention**. Gradual bladder decompression has no role — it does not reduce haematuria or diuresis.
- **First-line: free access to oral fluids**, without routine intravenous supplementation; most cases resolve with expectant management.
- **Pathological diuresis (rare)**: hypovolaemia and disturbed sodium, potassium, phosphate, calcium and magnesium → **metabolic panel every 4–6 hours** with careful replacement.

- **Patient unable to drink** (obtunded, sedated, postoperative) → **replace about 75% of urine volume with 0.45% (half-normal) saline.**
- **No approved drug therapy;** PDE5 and COX-2 inhibitors are supported only by animal and cell-culture data.

Renal Recovery

- **Recovery depends on both the duration and the degree of obstruction.**

CLASSIC CANINE OBSTRUCTION EXPERIMENTS

Study	Model	Recovery of function after relief
Kerr, 1954	Complete unilateral ureteral obstruction (UUO) for 1 week	About 60% of GFR recoverable
Vaughan and Gillenwater, 1971	Complete UUO of increasing duration	7 days → 100% (by 2 weeks after relief); 14 days → 70%; 4 weeks → 30%; 6 weeks → none
Leahy et al., 1989	Partial unilateral obstruction up to 60 days	14 days → complete; 28 days → 31%; 60 days → 8%; dominant lesion arteriolar vasoconstriction

- **Discrepancy at 1 week:** the two complete-UUO models differ (about 60% versus full recovery); the source does not reconcile them — the consistent message is **near-full recovery within 1–2 weeks, steep decline by 4 weeks, none by 6 weeks.**
- **Vascular basis (Fig. 90.25):** scanning electron microscopy of glomerular microvascular casts after obstruction shows **capillary collapse and irregularity** against the normal tuft — **postobstructive dysfunction is driven chiefly by vascular responses.**
- **Clinical translation:** most stone formers with hydronephrosis from a **small ureteral stone causing low-grade obstruction may be safely observed for 2–4 weeks** after the first colic, from a functional standpoint.
- **Beyond 4 weeks, human outcome is poorly defined and unpredictable;** full recovery has been reported up to **5 months** after relief of unilateral obstruction.
- **Factors favouring recovery:** lesser degree of obstruction, greater collecting system compliance, more pyelolymphatic backflow, younger age, **thicker renal cortex.**
- **Because these cannot be judged during the acute event, keep a low threshold to treat the obstruction or temporise with a stent or nephrostomy,** avoiding permanent tubular injury and fibrosis.

Concept — *Obstruction injures the kidney through its microvasculature before its tubules, and recovery tracks how long arteriolar vasoconstriction has lasted; hence the clock — weeks, not the degree of dilatation — governs when a urologist must intervene.*